

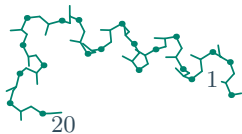
Sequence

L	T	F	C	E	I	I	L	M	E
M	E	S	H	D	A	A	W	P	F

20-residue excerpt

5-bit amino acid
representation

Backbone



N, C α , C, O

Frozen encoder + LFQ

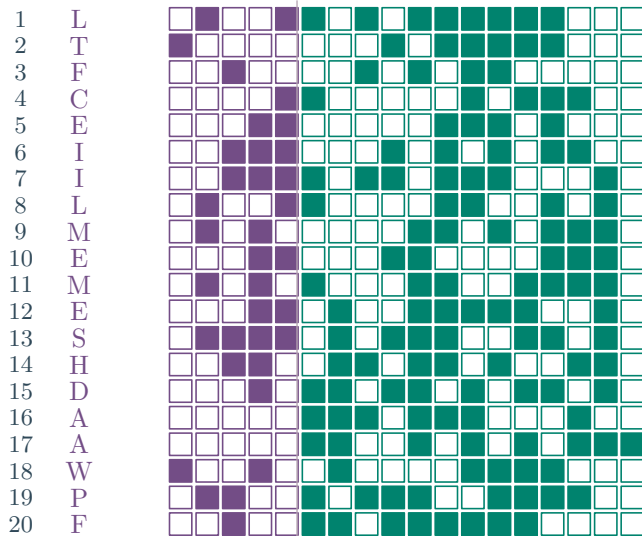
$$z \in \mathbb{R}^{13} \rightarrow b = \mathbf{1}[z > 0]$$

$$2^{13} = 8192 \text{ structural codes}$$

Residue

5 bits

13 LFQ bits



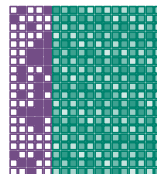
$$X_0 \in \{0, 1\}^{20 \times 18}$$

one row per residue

LFQ signs: $1 \mapsto +1, 0 \mapsto -1$

Forward folding

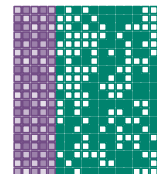
seq. struct.



fixed noisy

Inverse folding

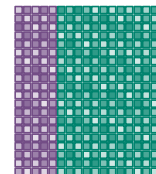
seq. struct.



noisy fixed

Joint generation

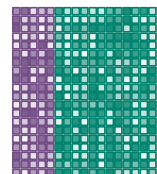
seq. struct.



noisy noisy

Noisy state

Y_k



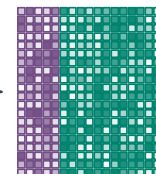
Denoiser

σ_k

D_θ

Prediction

P_k



DDIM update

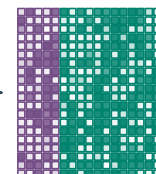
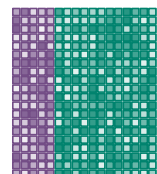
$$Y_{k+1} = P_k + \frac{\sigma_{k+1}}{\sigma_k}(Y_k - P_k)$$

Y_{k+1}

σ_{k+1}

D_θ

P_{k+1}



$$\sigma_{k+1} < \sigma_k$$

$\vdots$